Three-Fingered Microgripper for Medical Applications

Structural and Nonlinear Static Analysis

Author Alessio Chirichiello

Course Case Studies in Scientific Computing

Tools Autodesk Inventor ANSYS Mechanical R1 2025

3D Modelling

The design of the robotic hand comprises three fingers, each with two phalanges, connected via revolute joints to a central palm. An additional revolute joint in the thumb provides an extra degree of freedom. In total, the system features **seven joints** and **seven degrees of freedom**. Autodesk Inventor was used for 3D CAD modelling of each component and the full hand assembly, as shown in Fig. 1.

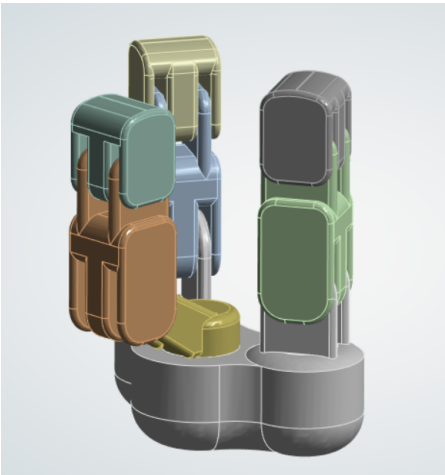


Figure 1. CAD model of the three-fingered gripper (Autodesk Inventor).

The dimensions are based on the **HIT Hand II** model from DLR, scaled by a factor of five. The initial design was then optimised to reduce total mass by restructuring the base, reducing it from **10.588 g** to **3.5905 g**. The phalange dimensions and masses are summarised in Table 1.

Table 1. Three-Fingered Gripper Geometric Parameters

Component	Dimensions (mm)	Mass (g)
Proximal Phalanx	$10 \times 4 \times 4$	0.6679
Distal Phalanx	$5 \times 4 \times 4$	0.3625

The hand is fabricated from **Stainless Steel 316L**, a low-carbon alloy widely used in medical devices and implants due to its excellent biocompatibility and corrosion resistance in biological fluid environments. Material properties are listed in Table 2.

Table 2. Stainless Steel 316L Material Properties

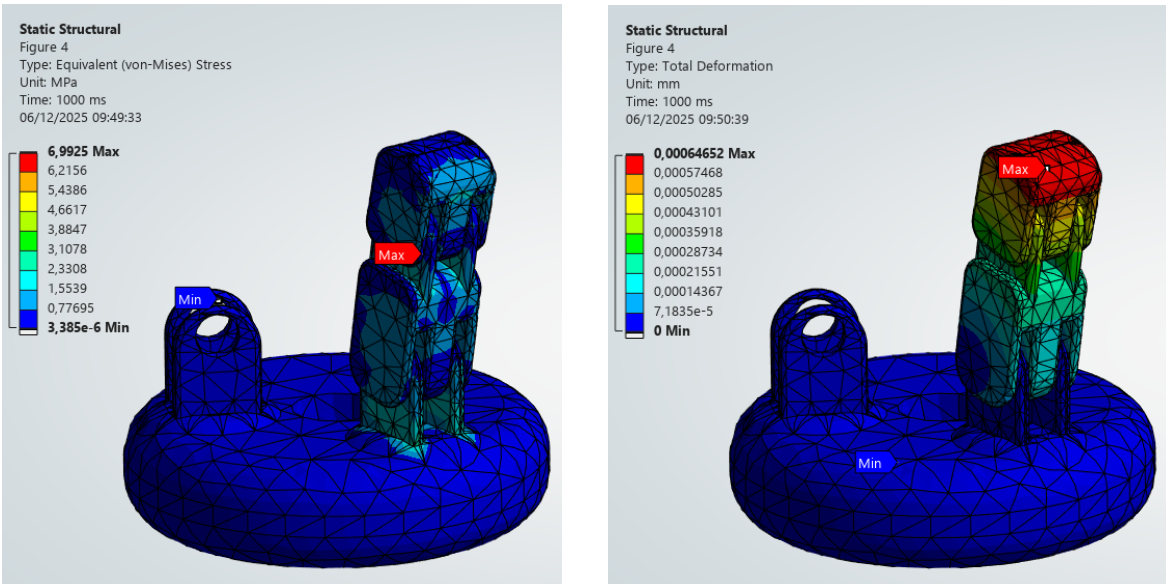
Property	Value	Unit
Young's Modulus (E)	190	GPa
Poisson's Ratio (ν)	0.3	—
Yield Strength (σ_y)	206	MPa

Linear Static Analysis

A linear static analysis is performed on the gripper to evaluate its structural integrity using **ANSYS Mechanical R1 2025**. Two modelling assumptions are made:

- Contact surfaces are *bonded* or not permitted to separate.
- Torsional stiffness of the revolute joints is neglected.

The boundary conditions consist of a fixed support at the gripper base to prevent rigid body motion. A **10 N** load is applied at each fingertip.



(a) Equivalent von-Mises stress. **(b)** Total deformation.

Figure 2. Linear static analysis results (ANSYS Mechanical).

Key results — Linear Static Analysis

Maximum equivalent von-Mises stress: **6.9925 MPa**
Maximum total deformation: **0.00064 mm**

The yield strength criterion is satisfied: $\sigma_{max} = 6.99\,\text{MPa} \ll \sigma_y = 206\,\text{MPa}$. The results were validated against a vertical cantilever bar under identical loading and boundary conditions.

Nonlinear Static Analysis

To perform a more physically realistic analysis, several improvements are made to the model.

Joint Stiffness Modelling

Joint compliance affects force transmission and must be accounted for. The torsional stiffness of each joint pin (modelled as a cylinder) is computed as:

$$K_t = \frac{G\,I_p}{L} \tag{1}$$

where G is the shear modulus (obtained at 20 °C by interpolating ANSYS tabular data), I_p is the polar moment of inertia, and L is the pin length. The resulting stiffness values are:

- **Standard joints** (proximal–distal and finger–palm connections): $K_t = 35{,}453$ N mm/rad
- **Thumb joint** (different geometry): $K_t = 321{,}440$ N mm/rad

Both values indicate quasi-stiff behaviour.

Bearing Model

As an alternative to direct joint stiffness, a **bearing condition** is applied directly at the pin-hole interface in ANSYS, without modifying the geometry. The bearing is modelled as a 2D elastic element characterised by the stiffness matrix:

$$\begin{bmatrix} F_1 \\ F_2 \end{bmatrix} = \begin{bmatrix} K_{11} & K_{12} \\ K_{21} & K_{22} \end{bmatrix} \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} \tag{2}$$

Cross-coupling terms are neglected and the matrix is assumed isotropic. The radial stiffness is estimated using El-Sayed's empirical formula for deep-groove ball bearings:

$$K = L\,W^{1/3} \tag{3}$$

where L is a geometry-dependent parameter and W is the applied load.

Frictional Contact and Results

The nonlinear analysis uses frictional contact between joints and components. For steel-on-steel interaction, the static coefficient of friction is $\mu_s = 0.6$ and the dynamic coefficient is $\mu_d = 0.3$. The bearing radial stiffness is set to $K = 1000$ N/mm and a load of **5 N** is applied at the fingertips.

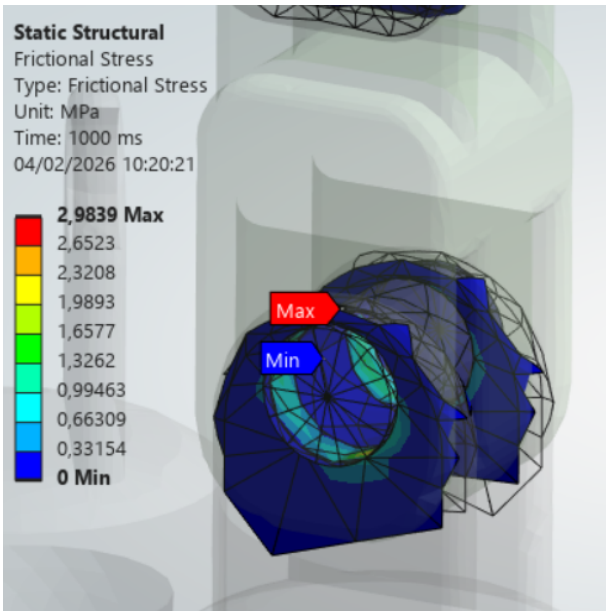


Figure 3. Frictional stress at bearing interfaces (nonlinear static analysis).

Key result — Nonlinear Static Analysis

Maximum frictional stress at bearing interface: **2.9839 MPa**

Grasp Stability Analysis

Grasping is a fundamental component of robot manipulation, requiring the application of forces and torques at contact points to fully constrain an object's motion. Once a grasp is performed, its robustness must be evaluated.

An additional object is modelled in AutoCAD and imported alongside the gripper into ANSYS Mechanical. The setup is as follows:

- Gripper fingers: *fixed*.
- Grasped object: *unconstrained* (stainless steel 316L).
- Contact between object and fingertips: *frictional*.
- Loading conditions: standard Earth gravity + incremental force up to **200 N** applied to one end of the object.

The maximum equivalent von-Mises stress remains below the material yield strength under all loading conditions.

Contact Status

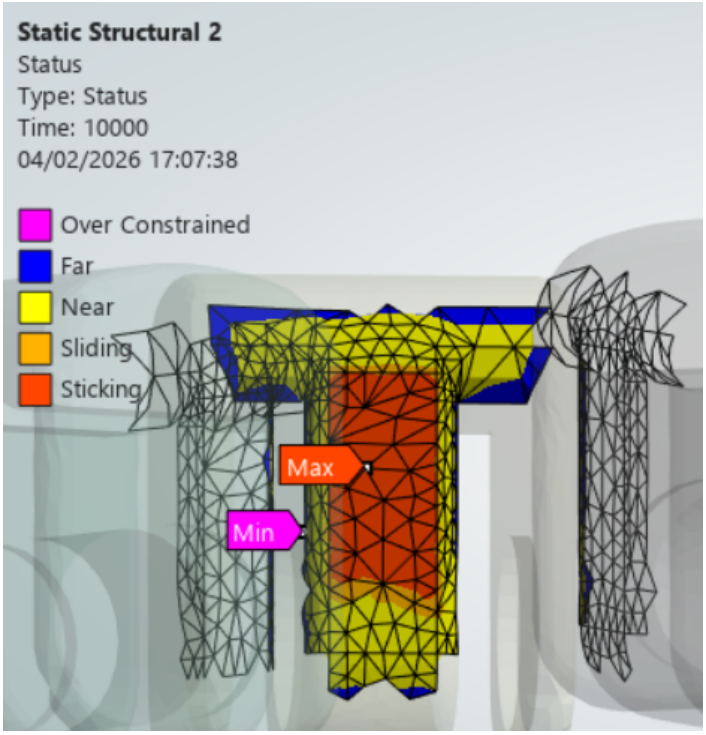
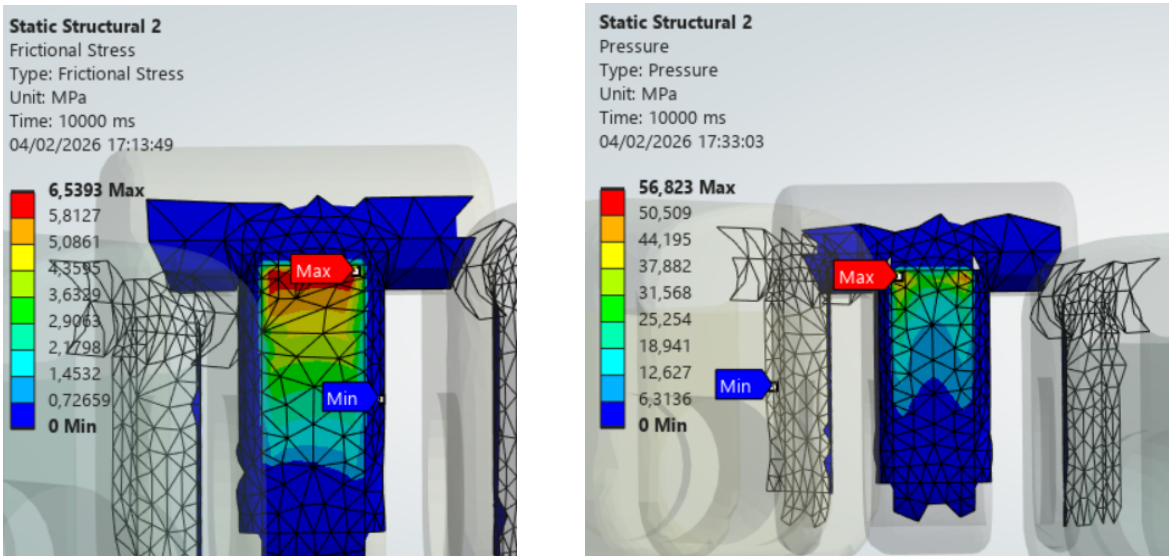


Figure 4. Contact status at fingertip–object interface.

As shown in Fig. 4, the contact status indicates a **sticking** condition throughout the simulation, confirming that the object does not slide within the grasp.

Coulomb Friction Verification

Grasp stability is verified by comparing frictional stress against the Coulomb friction limit $F = \mu\, p_{\max}$:



(a) Frictional stress at contact. **(b)** Contact pressure.

Figure 5. Grasp stability verification — frictional stress and contact pressure.

Grasp stability verification

Maximum frictional stress: **6.5 MPa**
Maximum contact pressure: **56.8 MPa**
Coulomb friction limit ($\mu \cdot p_{\max}$): **33.6 MPa**

Since $\tau_{\max} = 6.5$ MPa $< F_{\text{Coulomb}} = 33.6$ MPa, the grasp is confirmed to be **stable**.